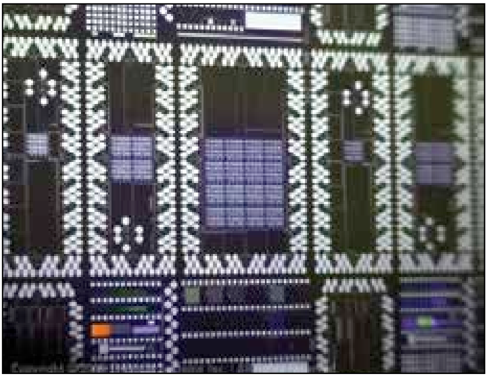
How a quantum computer would beat the living hell out of every encryption algorithm in the world

The science of the process of data encryption is known as cryptography. It is among the hot fields of research among scholars and corporates alike, as it pretty much has been. Among the reasons is how a downright simple method (to our awesome cognitive senses) is so freakishly hard for a computer to understand. As we know, 128 or 256 bit is considered sufficient in view of today's threats. The reasons for this were just examined by us. But recently, there have come into existence theoretical methods that threaten to hold current encryptions by the scruff of the neck, potentially packing havoc for all the 'belief' that online transactions have garnered after years of rather faithful service. Now, let us give you a basic idea of how exactly encryption works. The main principle of working here is the exponentially difficult nature of some problems in computer science. A common example is the factoring problem – where a number needs to be divided into its prime factors. While it may seem easy for you to do, but then, as we'd told you some time back, you are probably one of the foremost supercomputers in existence. For the run-of-the-mill, everyday hackers, we don't expect them to go anywhere beyond mainframes. And for systems of such limited power, the problem of factoring is notoriously difficult. So difficult, in fact, that they call it one of the 'hard' problems in computer science (no seriously, that is the word – hard problems). And with that, the difficulty in solving increases exponentially with every new bit (which is why we talked about 2256 when talking about 256-bit encryptions). With every new bit, the difficulty increases by a factor of 2.



But all of this can potentially change, if we can somehow pull a quantum computer out of the bag. They are revolutionary devices, which would pretty much change the fact of computing as we know it. The exact and detailed explanation for this is really, really difficult to cough up in this tiny space